

Find the value of one ratio part

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: In the ratio 4:7, the smaller part is 20. What is the larger part? Copy or complete the first step.

2. Share £420 in the ratio 3:4.

3. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: In the ratio 4:7, the smaller part is 20. What is the larger part? Copy or complete the first step.
Answer 35.
2. Share £420 in the ratio 3:4.
Answer £180 and £240.
3. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
Answer 52 km.
4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.
Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.

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Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. In the ratio 4:7, the smaller part is 20. What is the larger part?

6. Check this result using an inverse, estimate or known fact: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

2. Share £420 in the ratio 3:4.

7. Prove or explain your reasoning: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

3. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.

4. Solve this independently and show every stage: In the ratio 4:7, the smaller part is 20. What is the larger part?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Share £420 in the ratio 3:4.

10. Write and solve a similar question to: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

Teacher answers and checking prompts

1. In the ratio 4:7, the smaller part is 20. What is the larger part?
Answer 35.
2. Share £420 in the ratio 3:4.
Answer £180 and £240.
3. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
Answer 52 km.
4. Solve this independently and show every stage: In the ratio 4:7, the smaller part is 20. What is the larger part?
Answer 35.
5. Use a second method or representation for: Share £420 in the ratio 3:4.
Answer Expected result: £180 and £240. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
Answer Expected result: 52 km. A valid check must be shown.
7. Prove or explain your reasoning: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
Answer 24.5 cm. The explanation must show why the method works.
8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.
Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

6. Create a new example that tests the same idea as: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

2. Prove the answer to this question, rather than only calculating it: In the ratio 4:7, the smaller part is 20. What is the larger part?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Share £420 in the ratio 3:4.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 52 km.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
Answer 24.5 cm. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: In the ratio 4:7, the smaller part is 20. What is the larger part?
Answer Expected result: 35. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Share £420 in the ratio 3:4.
Answer Expected result: £180 and £240. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 52 km.
Answer One valid reconstruction is: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.
Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.
6. Create a new example that tests the same idea as: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.